JEE Mathematics: Quick Review with Grading and Practice Map - Monotonicity (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (59 questions; a question that appears in two of these is counted once): High = 5 or more questions, Medium = 1 to 4 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Monotonic function

Question 01

Define monotonic function

Model Answer

A function that is only increasing or only decreasing in its whole domain is monotonic; otherwise it is non-monotonic.

Ex: e^x, $\ln x$, x^3 are monotonic; x^2, $\sin x$ (on $\mathbb{R}$) are not.

Grade - Low

Practice map, main concept: DPP: 3(6)

Also used in: none

Question 02

Define monotonic increasing, monotonic decreasing function

Model Answer

Increasing: moving left to right the graph rises ($x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$). Decreasing: the graph falls ($x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$).

IIT language: increasing = strictly increasing; NCERT 'increasing' = non-decreasing.

Grade - Low

Practice map, main concept: none

Also used in: none

Monotonicity at a point

Question 03

What is the condition for a function to be increasing at $x = a$?

Model Answer

$f(a-h) < f(a) < f(a+h)$ for all sufficiently small $h > 0$

Sufficient test: $f'(a) > 0$. If $f'(a) = 0$ check the sign of f' on both sides (e.g. x^3 is increasing at 0).

Grade - Low

Practice map, main concept: DPP: 3(9)

Also used in: none

Question 04
What is the condition for a function to be decreasing at $x = a$?

Model Answer
$f(a-h) > f(a) > f(a+h)$ for all sufficiently small $h > 0$
Sufficient test: $f'(a) < 0$. Ex: $f'(x) = g(x)(x-\alpha)^2$ with $g(\alpha) < 0$: f is decreasing near α.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 05
What to keep in mind while checking increasing/decreasing if one side of $x = a$ is not in domain?

Model Answer
Compare only with the side that lies in the domain: at a left end point a, increasing means $f(a) < f(a+h)$; at a right end point b, $f(b-h) < f(b)$.
Ex: $\sqrt{x}$ is increasing at $x = 0$ (only $h > 0$ is checked).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 06
What is the condition on continuity of a function to define monotonicity?

Model Answer
None: monotonicity is defined only in the domain and the function need not be continuous (or differentiable).
Ex: $[x]$ is non-decreasing on $\mathbb{R}$; $-\frac{1}{x}$ is increasing on $(-\infty, 0)$ and on $(0, \infty)$ separately but not on their union.

Grade - Low
Practice map, main concept: none
Also used in: none

Monotonicity in an interval

Question 07
What is the condition for a function to be increasing in an interval?

Model Answer

$$\forall\, x_1, x_2 \in I :\ x_1 > x_2 \Rightarrow f(x_1) > f(x_2)$$

Equivalently $\dfrac{f(x_1) - f(x_2)}{x_1 - x_2} > 0$ for all $x_1 \neq x_2$ in I.

Grade - Low
Practice map, main concept: Sheet: C-I 13
Also used in: none

Question 08
What is the condition for a function to be decreasing in an interval?

Model Answer

$$\forall\, x_1, x_2 \in I :\ x_1 > x_2 \Rightarrow f(x_1) < f(x_2)$$

Ex: $\sin x$ is decreasing in $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 09
What is the condition for a function to be non-decreasing in an interval?

Model Answer

$$\forall\, x_1, x_2 \in I :\ x_1 > x_2 \Rightarrow f(x_1) \geq f(x_2)$$

The value never decreases (it may stay constant on a part). Ex: $[x]$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-II 122; J-M 48

Question 10
What is the condition for a function to be non-increasing in an interval?

Model Answer

$$\forall\, x_1, x_2 \in I :\ x_1 > x_2 \Rightarrow f(x_1) \leq f(x_2)$$

The value never increases. Ex: $-[x]$.

Grade - Low
Practice map, main concept: none
Also used in: none

Derivative approach for monotonicity

Question 11
What is the condition on derivative if $f(x)$ is increasing in an interval?

Model Answer
$f'(x) \geq 0 \ \ \forall x \in I$, equality not holding on any sub-interval
$f' = 0$ at isolated points is allowed: x^3, $x + \sin x$ are increasing on $\mathbb{R}$. If $f' \equiv 0$ on an interval, f is constant there.

Grade - Low
Practice map, main concept: Sheet: J-II 107, 122 | DPP: 3(10)
Also used in: Sheet: J-II 114

Question 12
What is the condition on derivative if $f(x)$ is decreasing in an interval?

Model Answer
$f'(x) \leq 0 \ \ \forall x \in I$, equality not holding on any sub-interval
Ex: $\cot^{-1} x$ ($f' = -\frac{1}{1+x^2} < 0$) is decreasing on $\mathbb{R}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 13
How to find intervals of monotonicity?

Model Answer
Find the domain, factorise $f'(x)$, mark its zeros and the points where it does not exist (and points outside the domain) on the number line, read the sign of f': $+$ increasing, $-$ decreasing.
Ex: $f = x^3 - 3x$, $f' = 3(x-1)(x+1)$: increasing on $(-\infty, -1] \cup [1, \infty)$, decreasing on $[-1, 1]$.

Grade - High
Practice map, main concept: Sheet: J-I 24-26, 28-29, 32, 50; J-II 28-29, 77-78, 89, 96-97, 99, 105, 112-114, 116, 124; J-M 3, 5, 28, 50, 52; J-A 2.01, 2.06-2.08, 2.11, 3.01; C-I 11, 17, 27, 33; C-II 17, 52 | DPP: 3(1-3, 5, 7-8, 11); 4(1-3); 6(12) | Question Bank: 155-156, 160, 162-163, 165-167
Also used in: Sheet: J-I 27; J-II 30, 90, 115, 127; J-M 2, 49; J-A 3.12; C-I 7 | Question Bank: 154, 158, 164

Question 14
How to check whether a boundary point will be included in interval of increasing/decreasing?

Model Answer
Include the point if it is in the domain and f is continuous there (a zero of f' or an end point of a closed domain). Exclude points not in the domain; at a jump compare the values directly (item 15).
Ex: $2x^2 - kx + 5$ increasing in $[1, 2]$: need $f'(1) = 4 - k \geq 0$, so $k \leq 4$ (equality at the end point is allowed).

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-I 28, 30; J-M 15

Question 15
How to find intervals of monotonicity for non-uniform function?

Model Answer
Find the monotonicity of each piece in its own interval; at a junction a compare $f(a^-)$, $f(a)$, $f(a^+)$ directly (the derivative may not exist there).
Ex: $f = x$ for $x < 1$, $f = 3 - x^2$ for $x \geq 1$: increasing on $(-\infty, 1]$ because $f(1^-) = 1 < f(1) = 2$, decreasing on $[1, \infty)$.

Grade - Medium
Practice map, main concept: Sheet: J-I 31; J-II 30, 104, 119, 127; J-M 2, 49; J-A 3.12; C-I 7 | Question Bank: 154
Also used in: Sheet: J-I 34-35; J-II 53

Question 16
What will be monotonic nature of $f(g(x))$ if monotonic nature of f and g is known?

Model Answer
Same nature ($\uparrow\uparrow$ or $\downarrow\downarrow$): $f(g(x))$ increasing. Opposite nature: $f(g(x))$ decreasing (because $(f \circ g)' = f'(g)\, g'$).
Ex: e^{-x} ($e^t \uparrow$, $-x \downarrow$) is decreasing; $\tan^{-1}(\sin x + \cos x)$ increases exactly where $\sin x + \cos x$ increases.

Grade - Medium
Practice map, main concept: Sheet: J-I 27
Also used in: Sheet: J-M 3, 31

Critical points and stationary points

Question 17
Define critical points

Model Answer
Points of the domain where $f'(x) = 0$ or $f'(x)$ does not exist.
Ex: $x^{2/3}$: $f' = \frac{2}{3x^{1/3}}$, critical point $x = 0$; $|x|$: $x = 0$; $\tan x$, e^x: none; $\{x\}$: every integer; $[x]$: every real x.

Grade - Medium
Practice map, main concept: Sheet: J-I 33-35, 37; J-II 35 | Question Bank: 157, 169
Also used in: Sheet: J-II 33-34; J-A 3.01

Question 18
Define stationary points

Model Answer
Points of the domain where $f'(x) = 0$ (tangent horizontal). Every stationary point is critical, not conversely.
Ex: x^3: $x = 0$ is stationary but f keeps increasing; $x + \sin x$: stationary at $x = (2n+1)\pi$.

Grade - Low
Practice map, main concept: Sheet: J-I 36
Also used in: Sheet: J-II 107

Always increasing/decreasing based

Question 19
How to find unknown in a function if it is always increasing/decreasing? Both methods

Model Answer
Method 1 (separate the parameter): $f'(x) \geq 0\ \forall x \Rightarrow$ parameter $\geq$ maximum of the other side (for decreasing: $\leq$ minimum).
Method 2 (f' quadratic in x or in $\sin x$, e^x...): use sign / location of roots of the quadratic ($a > 0$, $D \leq 0$ for $\mathbb{R}$). 'No critical points' means $f' \neq 0$ everywhere: strict inequality.
Ex: $ax - \sin x$ always increasing $\Leftrightarrow a - \cos x \geq 0\ \forall x \Leftrightarrow a \geq 1$; always decreasing $\Leftrightarrow a \leq -1$.

Grade - Medium
Practice map, main concept: Sheet: J-I 30; J-II 33-34, 36-38; J-M 15, 48 | DPP: 3(4, 13-14)
Also used in: Sheet: J-I 37; J-II 35, 83; J-M 30 | DPP: 3(12) | Question Bank: 169

Question 20

How to check whether a cubic polynomial is monotonic?

Model Answer

$f = ax^3+bx^2+cx+d$ is monotonic on $\mathbb{R} \iff f' = 3ax^2+2bx+c$ keeps one sign $\iff b^2-3ac \le 0$

$a > 0$: increasing, $a < 0$: decreasing. If $b^2 - 3ac > 0$ it has a local max and a local min (non-monotonic). Ex: $x^3 + x^2 + x$: $1 - 3 < 0$, increasing.

Grade - Medium

Practice map, main concept: Sheet: J-II 83-84; J-M 30 | DPP: 3(12)

Also used in: Sheet: J-I 36 | Question Bank: 159

Greatest and Least value

Question 21

How to find greatest and least value of a function in an interval with the help of intervals of monotonicity?

Model Answer

Find the intervals of monotonicity inside the given interval; compare f at the end points and at the critical points where the nature changes. A monotonic f on $[a, b]$ has its extreme values at a and b.

Ex: $x^3 - 3x$ on $[0, 2]$: $f(0) = 0$, $f(1) = -2$, $f(2) = 2$; greatest 2, least -2.

Grade - Medium

Practice map, main concept: Sheet: J-II 43-45, 56, 115; J-M 32; C-I 22, 30 | DPP: 4(13) | Question Bank: 172

Also used in: Question Bank: 162

Question 22

Where will greatest/least value of a continuous function exist in an interval?

Model Answer

On a closed interval $[a, b]$ a continuous function attains both, either at a critical point inside or at an end point a, b.

On an open interval they may not exist: x on $(0, 1)$ has neither.

Grade - Medium

Practice map, main concept: Sheet: J-A 2.04; C-I 14, 34

Also used in: Sheet: J-II 43, 56

Question 23

How to check whether a^b is greater/less than b^a?

Model Answer

$$a^b \gtrless b^a \iff \frac{\ln a}{a} \gtrless \frac{\ln b}{b}, \qquad \left(\frac{\ln x}{x}\right)' = \frac{1-\ln x}{x^2}$$

$\frac{\ln x}{x}$ increases on $(0, e]$ and decreases on $[e, \infty)$: for $e \le a < b$, $a^b > b^a$. Ex: $3^4 = 81 > 4^3 = 64$; $e^\pi > \pi^e$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 24

How to find range of a cubic polynomial for an interval of x?

Model Answer

Find the roots of $f' = 0$ lying in the interval; the range is $[\min, \max]$ of f at those points and the end points (the cubic is continuous).

Ex: $x^3 - 3x$ on $[-2, 3]$: $f(-2) = -2$, $f(-1) = 2$, $f(1) = -2$, $f(3) = 18$; range $[-2, 18]$.

Grade - Low

Practice map, main concept: none

Also used in: none

Proving Inequality

Question 25

How to solve questions on proving inequality with the help of monotonicity?

Model Answer

To prove $f(x) > g(x)$ for $x > a$: put $h = f - g$, show $h(a) \geq 0$ and $h'(x) > 0$ for $x > a$ (if the sign of h' is unclear, study h'' in the same way).

Ex: $h = x - \sin x$, $h' = 1 - \cos x \geq 0$, $h(0) = 0 \Rightarrow x > \sin x$ for $x > 0$; similarly $\tan x > x$ on $\left(0, \frac{\pi}{2}\right)$, $\ln(1 + x) < x$ for $x > 0$.

Grade - Medium

Practice map, main concept: Sheet: J-I 38-40; J-II 39-41, 75; C-II 18, 24 | Question Bank: 164

Also used in: Sheet: J-A 2.01; C-I 11; C-II 17 | Question Bank: 156

Rolle's Theorem

Question 26

Write statement of Rolle's theorem

Model Answer

If f is (i) continuous on $[a, b]$, (ii) differentiable on (a, b) and (iii) $f(a) = f(b)$, then $f'(c) = 0$ for at least one $c \in (a, b)$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: C-I 14, 30; C-II 21

Question 27

How to verify Rolle's theorem?

Model Answer

Check the three conditions one by one, then solve $f'(c) = 0$ and show that a root lies in (a, b).

Ex: $x^3 - 6x^2 + ax + b$ on $[1, 3]$: $f(1) = f(3) \Rightarrow a = 11$, b any real; then $c = 2 \pm \frac{1}{\sqrt{3}}$.

Grade - Low

Practice map, main concept: DPP: 4(10)

Also used in: Sheet: J-II 46; C-II 26

Question 28
How to find c of Rolle's theorem?

Model Answer
Solve $f'(c) = 0$ and keep the roots that lie strictly between a and b.
Ex: $x^2 - 4x + 3$ on $[1, 3]$: $2c - 4 = 0$, $c = 2$.

Grade - Medium
Practice map, main concept: Sheet: J-M 13; C-II 26 | DPP: 4(6, 11)
Also used in: none

Question 29
What can you say about location and count of c of Rolle's theorem?

Model Answer
c lies in the open interval (a, b), never at an end point; there is at least one c and there can be more than one.
Ex: $\sin x$ on $[0, 4\pi]$: $c = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 30
How to check whether Rolle's theorem is applicable for any given function?

Model Answer
All three conditions must hold together; failure of any one (a jump, a corner like $|x|$, $[x]$ inside, or $f(a) \neq f(b)$) makes it not applicable.
The conditions are sufficient, not necessary: $f'(c) = 0$ may still happen when they fail. Ex: $|x|$ on $[-1, 1]$: not applicable (not differentiable at 0).

Grade - Low
Practice map, main concept: Sheet: J-II 101; C-I 12, 26
Also used in: none

Question 31
What is geometrical definition of Rolle's theorem?

Model Answer
A continuous smooth curve with equal ordinates at $x = a$ and $x = b$ has at least one point between them where the tangent is horizontal (parallel to the x-axis).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 32
What is alternate definition of Rolle's theorem?

Model Answer
Between any two consecutive roots of $f(x) = 0$ (f differentiable) there is at least one root of $f'(x) = 0$.
Ex: between two roots of $e^x \cos x = 1$ there is a root of $e^x \sin x = 1$ (apply it to $e^{-x} - \cos x$).

Grade - Low
Practice map, main concept: Sheet: J-I 44; J-II 102
Also used in: Sheet: J-I 41 | DPP: 4(4, 8)

Question 33

What can you say about number of solutions of $f'(x) = 0$ if $f(x) = 0$ has n solutions?

Model Answer

$f(x) = 0$ has n real roots $\Rightarrow$ $f'(x) = 0$ has at least $n-1$, $f''(x) = 0$ at least $n-2$ real roots

Ex: $(x-4)(x-5)(x-6)(x-7)$: $f' = 0$ has a root in each of $(4,5)$, $(5,6)$, $(6,7)$.

Grade - High

Practice map, main concept: Sheet: J-I 42; J-II 51; J-M 35; J-A 2.05, 2.12; C-II 22 | Question Bank: 161, 168

Also used in: Sheet: J-II 76; J-M 12, 53

Question 34

What can you say about number of solutions of $f(x) = 0$ if $f'(x) = 0$ has m solutions?

Model Answer

$f'(x) = 0$ has m real roots $\Rightarrow$ $f(x) = 0$ has at most $m+1$ real roots

Ex: $f'(c) = 0$ for exactly two values of $c \Rightarrow f(d) = 0$ has 1, 2 or 3 real roots (or none).

Grade - Low

Practice map, main concept: Sheet: J-II 106

Also used in: none

Question 35

How to check whether a question is of Rolle's theorem?

Model Answer

The question asks to show that an equation has a root in (a,b) / 'at least one c' with an expression in $f'(c)$, or gives equal values $f(a) = f(b)$, or asks the least number of roots of f', f'' or of an expression like $f'^2 + ff''$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 36

How to assume function (on which Rolle's theorem is to be applied) in question of Rolle's theorem?

Model Answer

Read the required equation as $g'(c) = 0$ and integrate: $f'(c) + kf(c) = 0 \to g = e^{kx}f$; $cf'(c) = f(c) \to g = \frac{f(x)}{x}$; $f'(c) = k\,\phi'(c) \to g = f - k\phi$; a polynomial $\sum a_i c^i = 0 \to g = \sum \frac{a_i x^{i+1}}{i+1}$.

Ex: $6a+4b+3c+3d = 0$: take $g = \frac{a}{4}x^4 + \frac{b}{3}x^3 + \frac{c}{2}x^2 + dx$, $g(0) = g(2) = 0 \Rightarrow ax^3 + bx^2 + cx + d = 0$ has a root in $(0,2)$.

Grade - High

Practice map, main concept: Sheet: J-I 41, 43, 45; J-II 46-50; C-II 19 | DPP: 4(4-5, 8)

Also used in: Sheet: J-I 44; J-II 51, 102; J-M 35; J-A 2.05, 2.12 | Question Bank: 161, 168

Lagrange's Mean Value Theorem (LMVT)

Question 37
Write statement of LMVT.

Model Answer

f continuous on $[a,b]$, differentiable on $(a,b) \Rightarrow \exists\, c \in (a,b):\ f'(c) = \dfrac{f(b)-f(a)}{b-a}$

Alternative form: $f(a+h) = f(a) + h\, f'(a+\theta h)$, $0 < \theta < 1$.

Grade - Medium
Practice map, main concept: Sheet: J-II 52; J-M 12, 53; C-II 25
Also used in: Sheet: J-A 2.09-2.10; C-I 10; C-II 22

Question 38
Prove LMVT with the help of Rolle's Theorem?

Model Answer

$$g(x) = f(x) - \frac{f(b)-f(a)}{b-a}\,x$$

g is continuous, differentiable and $g(a) = g(b)$; Rolle gives $g'(c) = 0$, i.e. $f'(c) = \frac{f(b)-f(a)}{b-a}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 39
How to verify LMVT?

Model Answer
Check continuity on $[a,b]$ and differentiability on (a,b); then solve $f'(c) = \frac{f(b)-f(a)}{b-a}$ and show a root in (a,b).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 40
How to find c of LMVT?

Model Answer
Solve $f'(c) = \frac{f(b)-f(a)}{b-a}$; keep the roots in (a,b). For a quadratic lx^2+mx+n, $c = \frac{a+b}{2}$.
Ex: $\ln x$ on $[1,e]$: $\frac{1}{c} = \frac{1}{e-1}$, $c = e-1$; x^2 on $[1,3]$: $c = 2$.

Grade - Medium
Practice map, main concept: Sheet: J-I 46, 48 | DPP: 4(7, 12)
Also used in: Sheet: J-M 13; C-II 25

Question 41
What can you say about location and count of c of LMVT?

Model Answer
$c \in (a,b)$ (never an end point); at least one, possibly several; $\theta = \frac{c-a}{b-a} \in (0,1)$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 42

How to check whether LMVT is applicable for any given function?

Model Answer

Both conditions must hold: continuity on $[a, b]$ (including the end points) and differentiability on (a, b); for a piecewise function check continuity and $f'(a^-) = f'(a^+)$ at each junction.

Ex: $x^{2/3}$ on $[-1, 1]$: not applicable (not differentiable at 0).

Grade - Low

Practice map, main concept: Sheet: J-II 53, 126 | DPP: 4(9)

Also used in: none

Question 43

What is geometrical definition of LMVT?

Model Answer

Between $A(a, f(a))$ and $B(b, f(b))$ there is at least one point C on the curve where the tangent is parallel to the chord AB.

Physically: the instantaneous rate equals the average rate at some instant.

Grade - Low

Practice map, main concept: none

Also used in: DPP: 4(12)

Question 44

How to solve questions of inequality with the help of LMVT by observing pattern $l < \dfrac{f(b) - f(a)}{b - a} < m$?

Model Answer

$m \le f'(x) \le M$ on $(a, b) \Rightarrow m(b - a) \le f(b) - f(a) \le M(b - a)$

Recognise the difference quotient, apply LMVT to get $f'(c)$, then bound $f'(c)$ using the monotonicity of f'. Ex: $f(1) = -2$, $f' \ge 2$ on $[1, 6] \Rightarrow f(6) \ge -2 + 2 \cdot 5 = 8$.

Grade - Medium

Practice map, main concept: Sheet: J-I 47; J-II 54-55, 125; J-A 2.09-2.10; C-I 10; C-II 20-21

Also used in: none

Plotting the graph

Question 45

How to plot graph of a function by finding limit, intervals of monotonicity?

Model Answer

Steps: domain; limits at the boundary points of the domain and at $\pm\infty$; intervals of monotonicity; values at critical points; points on the axes; asymptotes; then sketch.

Ex: $x^{1/x}$: domain $(0, \infty)$, $\to 0$ as $x \to 0^+$, $\to 1$ as $x \to \infty$, increasing on $(0, e]$, decreasing on $[e, \infty)$, greatest value $e^{1/e}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 103; C-II 47-48, 50

Also used in: Sheet: J-II 28, 77-78, 120; J-A 2.06-2.08; C-I 32 | DPP: 4(13)

Question 46

How to plot graph of a cubic polynomial?

Model Answer

Sign of a gives the ends ($a > 0$: from $-\infty$ to $+\infty$). Roots of $f' = 0$: if $b^2 - 3ac > 0$ there are a local max and a local min; their values fix how often the graph cuts the x-axis. If $b^2 - 3ac \leq 0$ the graph is monotonic.

Ex: $x^3 - 3x + 1$: max $f(-1) = 3 > 0$, min $f(1) = -1 < 0$, so three real roots.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: C-I 8

Question 47

How to plot graph of quadratic/quadratic?

Model Answer

Find the vertical asymptotes (real roots of the denominator), the horizontal asymptote $y = \frac{a_1}{a_2}$, the roots, the sign of f' (its numerator is a quadratic) and the turning values; then sketch. This also gives the range and one-one / onto nature.

Ex: $\dfrac{x^2 - x + 1}{x^2 + x + 1}$: $f' = 0$ at $x = \pm 1$, $f(1) = \frac{1}{3}$, $f(-1) = 3$, $y \to 1$; range $\left[\frac{1}{3}, 3\right]$.

Grade - Medium

Practice map, main concept: Sheet: J-II 90, 92-94

Also used in: Sheet: J-A 2.11 | Question Bank: 166

Asymptote

Question 48

What is the condition if $x = a$ is a vertical asymptote for $y = f(x)$?

Model Answer

$\lim\limits_{x \to a^+} f(x) = \pm\infty$ or $\lim\limits_{x \to a^-} f(x) = \pm\infty$ (at least one side)

Ex: $x = 0$ for $\ln x$; $x = \frac{\pi}{2}$ for $\tan x$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: C-I 29

Question 49

What is the condition if $y = a$ is a horizontal asymptote for $y = f(x)$?

Model Answer

$\lim\limits_{x \to \infty} f(x) = a$ or $\lim\limits_{x \to -\infty} f(x) = a$

Ex: $y = 0$ for e^x (as $x \to -\infty$); $y = \pm\frac{\pi}{2}$ for $\tan^{-1} x$.

Grade - Low

Practice map, main concept: Sheet: J-II 120; C-I 29

Also used in: none

Question 50

What is the condition if $y = mx + b$ is an asymptote for $y = f(x)$?

Model Answer

$$\lim_{x\to\pm\infty}\big(f(x)-(mx+b)\big)=0, \qquad m=\lim\frac{f(x)}{x}, \quad b=\lim\big(f(x)-mx\big)$$

Ex: $y = x + \frac{1}{x}$ has the inclined asymptote $y = x$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 51

How to find vertical asymptote for a rational function?

Model Answer

Cancel common factors; each real root a of the remaining denominator gives $x = a$.

Ex: $\dfrac{x^2+x+1}{(x+1)(x-2)}$: $x = -1$, $x = 2$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 52

What is the condition if horizontal asymptote exists for a rational function and how to find it?

Model Answer

$$f=\frac{a_0x^n+\dots}{b_0x^m+\dots}: \quad n<m\Rightarrow y=0; \qquad n=m\Rightarrow y=\frac{a_0}{b_0}$$

If $n > m$ there is no horizontal asymptote.

Ex: $\dfrac{2x^2+1}{x^2-4}$: $y = 2$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 92

Question 53

What is the condition if inclined asymptote exists for a rational function and how to find it?

Model Answer

Degree of numerator = degree of denominator +1. Divide: $f = (ax + b) + \frac{r(x)}{Q(x)}$; then $y = ax + b$ is the asymptote.

Ex: $\dfrac{x^3+x^2+x+3}{x^2-1} = x+1+\dfrac{2x+4}{x^2-1}$: $y = x + 1$.

Grade - Low

Practice map, main concept: none

Also used in: none

Miscellaneous

Question 54

How to find the number of real roots of $f(x) = 0$ (or the number of points where $y = f(x)$ cuts the x-axis) using monotonicity?

Model Answer

Split the domain at the critical points; on each interval f is monotonic, so it has a root there exactly when f changes sign between the end values (limits at $\pm\infty$ included). If $f' > 0$ everywhere there is exactly one root (for $f \to \mp\infty$). For e^x-polynomials put $t = e^x > 0$.

Ex: $3x^4 + 4x^3 - 12x^2 + 4$: $f' = 12x(x+2)(x-1)$, $f(-2) = -28$, $f(0) = 4$, $f(1) = -1$, $f \to \infty$ at both ends: 4 real roots. $x^7 + 5x^3 + 3x + 1$: $f' > 0$, one root.

Grade - **High**

Practice map, main concept: Sheet: J-II 42, 76; J-M 7, 11, 14, 29, 33-34, 51; J-A 2.02-2.03, 2.13; C-I 8, 32; C-II 23 | Question Bank: 153, 158-159, 170-171, 173

Also used in: none

Question 55

How to solve an inequality of the type $f(A(x)) > f(B(x))$ when f is monotonic?

Model Answer

$$f \uparrow:\ f(A) > f(B) \iff A > B; \qquad f \downarrow:\ f(A) > f(B) \iff A < B$$

First prove the monotonicity of f (sign of f'), rewrite the given inequality in the form $f(A) > f(B)$, then drop f (A, B must lie in the domain). Ex: $f = x^3 + x$ is increasing; $f(2a+1) > f(a+3) \iff 2a + 1 > a + 3 \iff a > 2$.

Grade - **Medium**

Practice map, main concept: Sheet: J-II 31-32; J-M 31; C-I 9

Also used in: none